DO NOW — RECALL

Input and output devices

Name: _____ Class: _____ Date: _____

1. Why is a GPU suitable for applying the same filter to many independent pixels?

2. What role does a processor's control unit have?

3. What data type should represent the state of a binary switch?

4. ASCII encodes A as decimal 65 and B as 66. A string 'ABBA' uses four 8-bit stored code units in this simplified example. How many bits and bytes, and why is this not proof all Unicode characters fit in one byte?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It can apply similar computations to many data elements in parallel.

2. Model answer

It decodes and coordinates instruction execution.

3. Model answer

Boolean.

4. Model answer

4×8=32 bits=4 bytes. Unicode contains many more characters, and UTF encodings may use more than one byte for a character; 8-bit storage here is an assumption.

Vocabulary — know these terms

sensor — Input transducer measuring a physical quantity.

ADC — Analogue-to-Digital Converter.

actuator — Output component that causes physical action.

latency — Delay between input and system response.

accessibility — Design usable by people with varied needs.

NEW LEARNING

Read, annotate and explain

Select suitable input

A temperature sensor measures a physical quantity; an analogue-to-digital converter is needed if the controller uses digital samples.

Process and control

Software compares the sampled temperature with a target to decide whether to open the vent.

Choose appropriate outputs

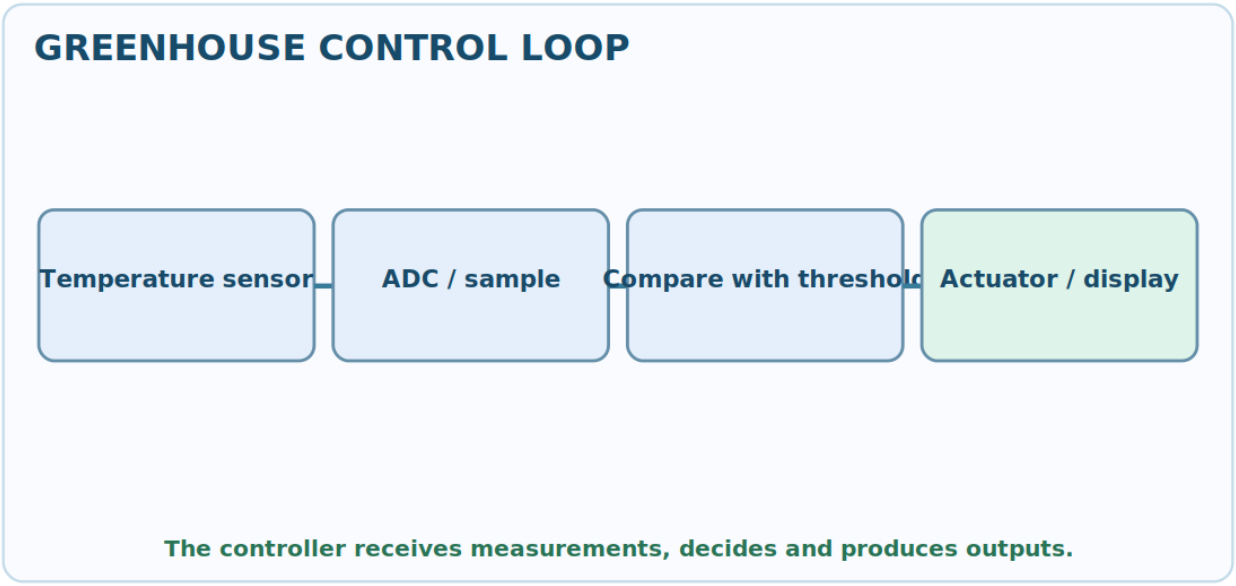
An actuator makes the physical change; a display can show readings. Design must address accessibility and suitable latency.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

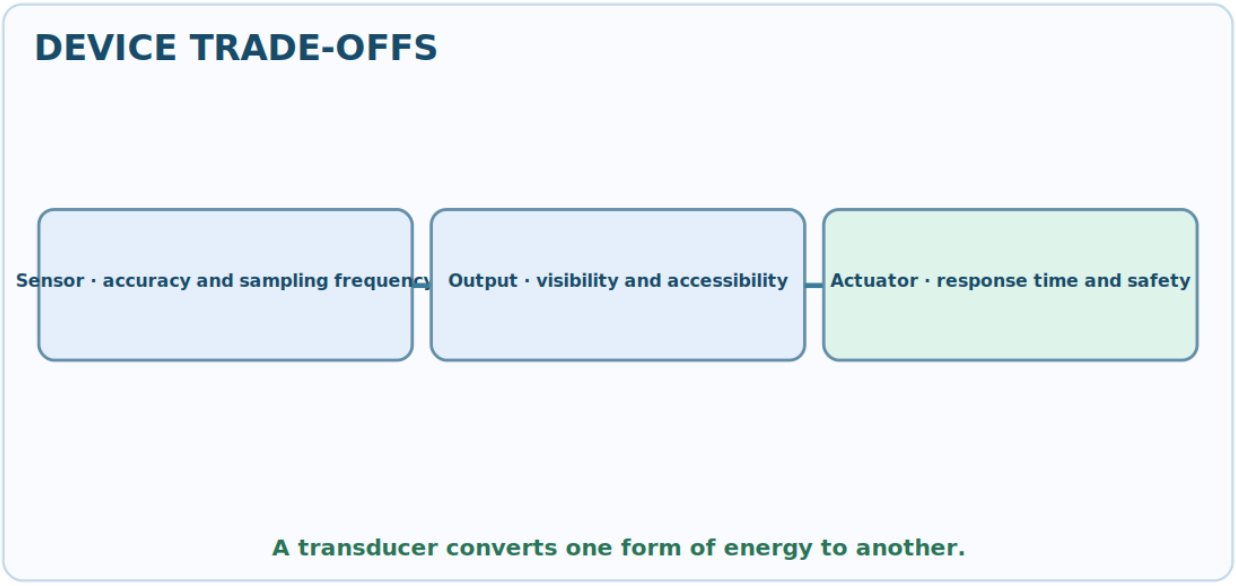
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



GREENHOUSE CONTROL LOOP: The controller receives measurements, decides and produces outputs.

New Learning



DEVICE TRADE-OFFS: A transducer converts one form of energy to another.

I DO — WORKED EXAMPLE

Follow the worked method

Choose input and output for an automatic door that opens when someone approaches.

Predict first

A proximity or motion sensor detects someone approaching and passes a signal to the controller.

Explain and check

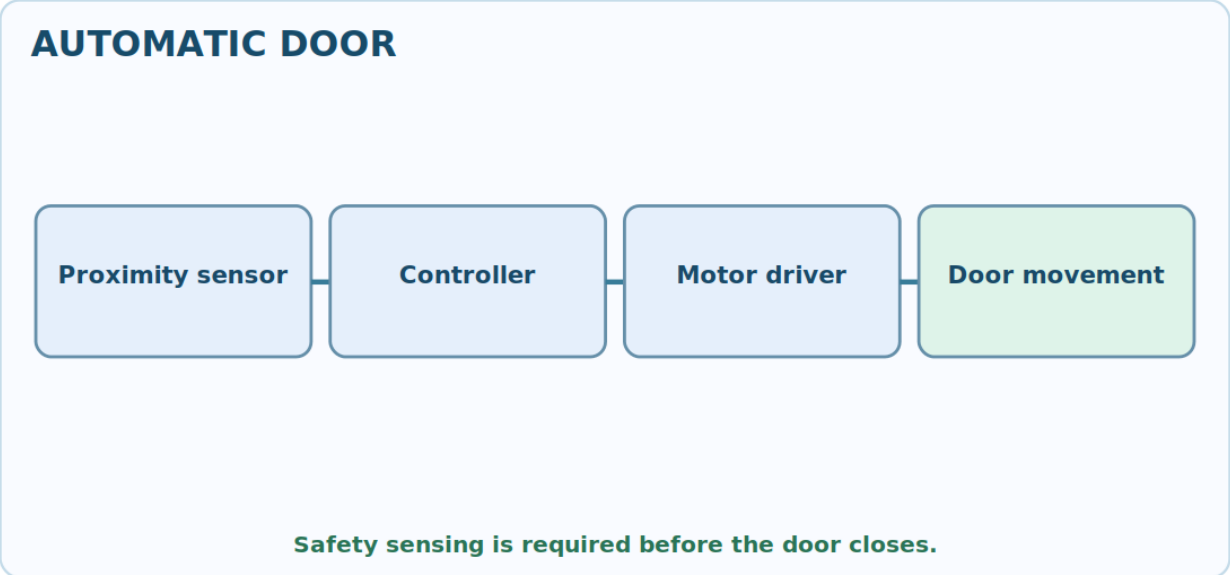
A motor and driver create movement; safety sensing reduces the risk of closing onto a person.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Choose input and output for an automatic door that opens when someone approaches.

A proximity or motion sensor detects someone approaching and passes a signal to the controller.



AUTOMATIC DOOR: Safety sensing is required before the door closes.

WE DO — GUIDED PRACTICE

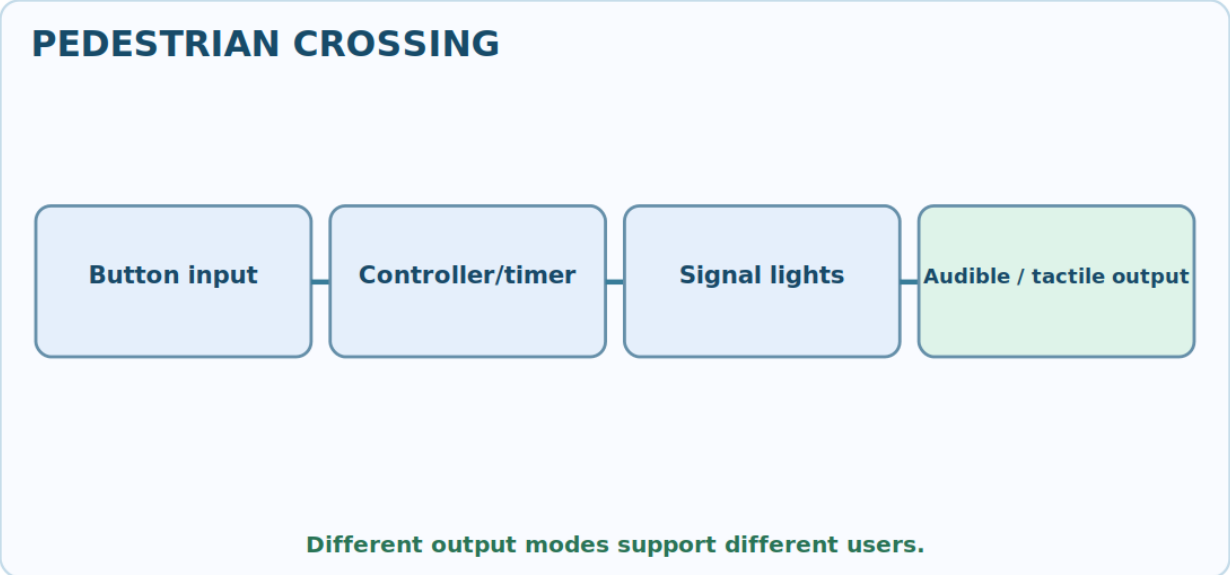
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Choose devices for an accessible pedestrian crossing with a push button, lights and audible alert.

A push button is a clear intentional input and the controller processes it under safe traffic timing rules.



PEDESTRIAN CROSSING: Different output modes support different users.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first A push button is a clear intentional input and the controller processes it under safe traffic timing rules.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Name a suitable input and output device for a weather station, and explain the role of each.

Question 2

A greenhouse opens a vent when temperature exceeds a limit. Name the sensor, processing decision and actuator, and explain any analogue conversion.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate device choices for a pedestrian crossing serving users with hearing or visual impairment.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A motor that physically opens a door is an input device.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain and check

Outputs include signal lights plus appropriate audible or tactile cues to improve accessibility.

IMPROVE: Explain one correction to your own reasoning.